Knowing When to Stop Asking: A Formal Coverage Criterion for Multi-Turn User Intent Understanding

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Abstract

Users hand conversational AI systems natural-language inputs too incomplete to drive downstream tasks without risk. We formalise user intent understanding as a *weighted coverage optimisation problem*. A domain parameter space $\mathcal{P}_d$ with importance weights and a coverage metric $\text{cov}(F) \in [0, 1]$ serves as both optimisation objective and stopping criterion, and a five-component pipeline interviews the user until coverage exceeds a domain-calibrated threshold θ. We validate across three deployments: RadAssist (clinical oncology), Telos (software project scoping) and FinAgent (autonomous financial agent constraint enforcement). The baselines establish the deficit: single-pass LLM extraction reaches 40.7% [37.5–43.9%] weighted coverage, and the strongest multi-turn LLM baseline (Claude Opus 4.8 active interview, $n = 150$) 58.4% [55.5–61.2%], under an LLM-as-judge scorer we validated against human annotations ($\kappa = 0.749$, $n = 516$). Phase 2 tests ALI on the same 150-prompt set: ALI reaches **89.0%** [88.2–89.9%] weighted coverage (+30.6 pp over the strongest frontier baseline, +26.0 pp when held to that baseline's eight-question budget), with 146/150 prompts stopped by the formal coverage criterion rather than a turn cap. The deficit is structural in the sense we can measure: at a fixed question budget it survives an order-of-magnitude increase in model capacity, and an explicit stopping criterion, rather than a larger model, is what moves it. The comparison favours ALI on two counts we declare and do not control, so the gap is an upper bound on what the criterion alone contributes.

1 Introduction

Users express less than a downstream system requires; for natural-language interfaces to structured AI pipelines, that gap is the default condition. A user who configures a trading agent with "keep it under $500 per trade and don't go crazy with the daily spending" has expressed risk posture while leaving unstated the exchange, the funding account and the weekly cap: parameters the agent must resolve before executing any financial action. Proceeding on an incomplete frame costs an unauthorised transaction, a miscoped engineering project, a corrupted prognostic model. Single-pass LLM extraction fails in three ways we name *incompleteness* (no measure of how much is understood), *passivity* (nothing probes for what is unstated) and *flatness* (every requirement weighs the same).

This paper. We reformulate user intent understanding as a *sequential weighted coverage optimisation problem*: for each domain, a parameter space $\mathcal{P}_d$ with importance weights and a coverage metric $\text{cov}(F) \in [0, 1]$ serve as the optimisation objective and stopping criterion. A five-component pipeline, the Adaptive Language Interview (ALI), selects at each turn the question that maximises expected coverage gain and terminates when coverage crosses a domain-calibrated threshold θ. This provides an explicit completeness criterion, relative to a curated parameter space, an active questioning mechanism, and a weighted prioritisation objective, none of which exist in single-pass extraction. We implement ALI and validate it across three separately built deployments. What is new is not the pipeline but the criterion: the interview is driven and

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terminated by an importance-weighted coverage function over an explicit parameter list, and the evaluation shows that this criterion, not model scale, is what moves coverage. Throughout, coverage is the share of importance-weighted parameters in an expert-written list that the conversation has pinned to a concrete value, as judged by a scorer we validate against human annotation; 89% is 89% of that list, not 89% task success. The gain comes at ≈1.3 times the tokens of the strongest baseline, a cost we quantify rather than assume, and two differences of protocol favour ALI that we declare rather than control (Section 5).

Contributions.

(1) A formal definition of user intent understanding as weighted coverage optimisation, with a measurable stopping criterion.
(2) A question-selection mechanism (semantic clustering with a reward optimised by genetic algorithm over ≈120,000 simulated episodes), reported as a design choice rather than a measured gain: a paired $n = 30$ ablation finds it equivalent to simpler orderings within pre-registered margins.
(3) A cross-domain stability signal: the same five-component decomposition required no architectural modification across three separately built deployments (clinical, software engineering, financial).
(4) An empirical evaluation ($n = 150$ prompts, validated LLM-as-judge scorer, $\kappa = 0.749$) quantifying the coverage gap: single-pass 40.7% [37.5–43.9%] and 58.4% for the strongest multi-turn baseline (Claude Opus 4.8), against 89.0% for ALI.
(5) Evidence that model capacity does not move the plateau: at a fixed question budget, three frontier models spanning an order of magnitude of capability land within 3 pp of one another (57.1–59.9%), while the same budget under an explicit coverage criterion yields 84.4%.

2 Related Work

Prior work relates to ALI along one axis: whether a system has a *measurable* criterion for when enough has been asked. The strands below each supply part of the machinery but leave that criterion implicit. Supplement § J[1] gives the full positioning, how the literature search was run and what would refute the claim.

Slot filling, dialogue state tracking and dialogue policy. Task-oriented dialogue gathers structured information through slot filling and DST [Young et al., 2013, Williams et al., 2016, Henderson et al., 2014]; mixed-initiative systems let both parties take initiative [Allen et al., 1999, Walker et al., 1997], and POMDP- or PPO-trained policies learn when to ask at the cost of millions of episodes [Young et al., 2013, Schulman et al., 2017]. This line leaves the termination condition to task-specific heuristics; we make it a weighted, measurable criterion.

Clarification questions and intent elicitation for LLM agents. Clarification work selects the single most informative question in retrieval settings [Rao and Daumé III, 2018, Majumder et al., 2021, Aliannejadi et al., 2020] or decides whether to clarify at all [Kuhn et al., 2022, Zhang et al., 2024], question by question; we decide against a weighted budget over the whole parameter space. Closest to our setting, Qian et al. [2024] train an agent to judge task vagueness before execution, Yao et al. [2024] evaluate agents against a simulated user under domain policies, and interview agents structure elicitation through an ontology of concerns [Jin et al., 2026a]; none ties termination to a measure of task coverage. A separate line gives clarification a *formal* stopping rule, but keys it to belief-state uncertainty or expected value of information [Suri et al., 2025, Choudhury et al., 2026] rather than to task coverage. These two lines do not meet: work that *measures* coverage does so offline, as an evaluation metric, while work that *stops* formally does so on uncertainty. We are not aware of prior work that makes an importance-weighted coverage function both the objective and the calibrated stopping criterion of an active interview.

Fine-tuning and evaluation. We fine-tune LoRA adapters [Hu et al., 2022] on 124 M–1.5 B models for structured sub-tasks; the evaluation finds they contribute nothing measurable. Hudeček and Dušek [2023] and Jin et al. [2026b] show that prompted LLMs resolve fewer than half of the implicit requirements of a multi-parameter task. Two methods our evaluation depends

[1] Supplementary material, released with the code as `supplement.pdf`: https://github.com/aonymous412/interview-pipeline-release/blob/main/supplement.pdf.

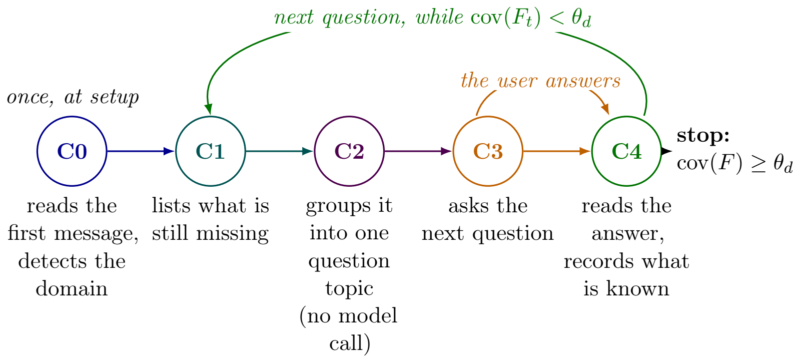

Figure 1: The ALI loop. C0 runs once on the initial message; C1–C4 run at every turn: C1 and C2 decide what to ask about, C3 asks it, the user answers, and C4 records the answer before the loop returns to C1. It closes only when weighted coverage crosses the domain threshold θ_d; the criterion sets interview length, rather than a preset budget. C2 is the only component that makes no model call.

on are themselves studied objects: the LLM judge [Zheng et al., 2023], which we control against human annotation, and the simulated user [Schatzmann et al., 2007], whose validity bounds what any oracle-based result can claim.

3 Method: The ALI Framework

3.1 Formal Problem Statement

Definition 1 (Parameter space, state frame and weighted coverage)**.** *For domain d, $\mathcal{P}_d = \{p_1, \ldots, p_n\}$ is a finite set of parameters with importance weights $w_d : \mathcal{P}_d \to \mathbb{R}^+$, power-law distributed by design (core $w \in [80, 100]$, secondary $[40, 70]$, optional $[10, 35]$). A* state frame *$F : \mathcal{P}_d \rightharpoonup \mathcal{V}$ is a partial assignment over $\mathcal{V} = \mathit{Strings} \cup \mathit{Numerics} \cup \mathit{Booleans} \cup \{\perp\}$, where $\perp$ denotes explicit user declination. Coverage is the share of* importance mass *the frame has pinned down, so resolving one high-weight parameter outweighs several trivial ones:*

$$\mathrm{cov}(F) \;=\; \frac{\displaystyle\sum_{p \in \mathrm{dom}(F)} w_d(p)}{\displaystyle\sum_{p \in \mathcal{P}_d} w_d(p)} \;\in\; [0, 1] \tag{1}$$

Coverage is monotonically non-decreasing as F grows. A parameter counts only once it resolves to a concrete value or an explicit declination; discussion without commitment does not extend $\mathrm{dom}(F)$ *(Section 4.3).*

Definition 2 (Coverage optimisation objective)**.** *Given domain d and threshold $\theta \in (0, 1)$, find the minimum-length question sequence such that:*

$$\min_{q_1, \ldots, q_T} \; T \quad \textit{s.t.} \quad \mathrm{cov}\big(F \mid q_1, \ldots, q_T\big) \;\geq\; \theta \tag{2}$$

In words: ask as few questions as possible while driving coverage past the threshold θ; the interview stops once enough of the important parameters are settled, whatever the turn count.

On "formal completeness." Equation (2) provides a coverage guarantee *relative to a manually curated $\mathcal{P}_d$*: once $\mathrm{cov}(F) \geq \theta$ holds, the system is θ-complete with respect to the defined taxonomy, and inherits the taxonomy's limits. Greedy selection (highest-weight uncovered parameter first) is in general suboptimal, since one question can address several related parameters at once.

3.2 The ALI Loop

Figure 1 shows the wiring of the five components; Algorithm F.1 in Supplement § F states the same loop precisely.

C0 (domain detector and pre-extractor): a keyword-weighted scorer identifies the active domain(s), and ≈15 regex pattern groups pre-extract facts from the initial message, initialising F_0. **C1 (element identifier)**: a fine-tuned LM outputs the full weighted parameter space for the detected domain, with a fuzzy name matcher and a knowledge-base fallback for low-confidence outputs; it re-runs at every turn against the current frame, so the uncovered set contracts. **C2 (semantic clusterer)**: groups uncovered parameters so that one question can address several related ones, and re-sorts clusters by total unresolved weight after each turn; simulation projected substantial turn savings from clustering, which the paired ablation below does not reproduce. **C3 (question generator)**: generates six candidate questions from a three-tier source (API LLM / local fine-tuned LM / expert template bank) and scores them with Eq. (3). **C4 (answer extractor)**: a fine-tuned LM parses user responses into structured key-value pairs, resolving targeted parameters and flagging *bonus elements* the user volunteers unasked; non-answer detection (≈30 phrases) prevents false positives.

3.3 Question Selection and Stopping Criterion

C3 scores each candidate question q at turn t by

$$R(q,t) = \underbrace{\sum_{e \in T(q)} w_e}_{\text{coverage gain}} + \underbrace{\beta\,|T(q)|\,\mathbf{1}[|T(q)|>1]}_{\text{multi-element bonus}} + \underbrace{\gamma\,\mathbf{1}[\text{cluster}(q)]}_{\text{cluster bonus}} + \underbrace{\delta\,\mathbf{1}[t<3 \wedge \bar{w}_{T(q)} > 70]}_{\text{early-turn bonus}} \quad (3)$$

where $T(q)$ is the set of parameters targeted by q and (β, γ, δ) are weights fitted by evolutionary search over ≈120,000 simulated episodes (Supplement § F gives the fitness function and the seven-phase training pipeline). The reward encodes three heuristics: maximise coverage gain per turn, prefer cluster-spanning questions, and front-load high-weight parameters while user cooperation is highest. Thresholds scale with domain complexity $|\mathcal{P}_d|$, from 0.75–0.85 for ten parameters or fewer to 0.95 above thirty, as starting points derived by Monte Carlo simulation; deployed thresholds may depart from these when weight mass is concentrated (Section 3.4). The ablation of Section 4.4 finds the fitted ordering equivalent to simpler ones within pre-registered margins, so we report the reward and its weights as a design choice rather than as a calibrated objective.

3.4 Three Instantiations

All three systems share the five-component architecture and differ only in domain knowledge base, base model size, threshold and enforcement mechanism (Supplement § E, Table E.1). All three are the authors' own systems, the financial one deployed in an applied context; their inclusion enables cross-domain comparison but is not independent third-party validation, and we anonymise the financial system's name for blind review.

Rad-Assist (Medical) runs at $\theta = 0.90$: any uncovered high-weight parameter ($w \geq 80$) can corrupt the downstream prognostic model. It uses a Mistral 7B model fine-tuned via GRPO [Shao et al., 2024] on MIMIC-CXR reports [Johnson et al., 2019]; in the evaluation below, C1/C4 run on Gemini 2.5 Flash, so the Medical figures characterise the architecture on a generic LLM, and the clinical model awaits its own evaluation with radiation oncologists. **Telos** (Consulting) maps developer descriptions to a structured `context.md` consumed by downstream code generators and estimators, over 15 project categories × 14–17 elements each (≈240 total), with θ scaling from 0.85 (simple landing page) to 0.95 (enterprise SaaS). **FinAgent** (Payments) adds an element the other two lack: strict separation of *probabilistic understanding* (ALI, once at setup) from *deterministic enforcement* (a validation engine, on every agent action). It uses two fine-tuned adapters, Qwen2.5-0.5B for C1 and Qwen2.5-1.5B for C4; the evaluation taxonomy below is a separate, coarser reconstruction over eleven parameters. It runs at $\theta = 0.75$, an operating point rather than a ceiling: the engine gates every later action, so pinning the highest-consequence constraints in 3–5 turns is the intended behaviour. Raising the threshold to 0.90 with everything else fixed lifts the domain to 93.9% in 5.32 turns (Supplement § C); the headline therefore understates rather than flatters.

4 Results

4.1 Setup and Protocol

We constructed 150 prompts, 50 per domain (Consulting/Telos, Medical/Rad-Assist, Payments/FinAgent), at three vagueness levels (ultra-vague $n = 43$; medium $n = 62$; detailed $n = 45$). An LLM-as-judge (Gemini 2.5 Flash) scores coverage; we validated it against 516 human annotations (Cohen's $\kappa = 0.749$ [Landis and Koch, 1977]). The scorer errs towards under-assigning RESOLVED, so the reported figures are, if anything, underestimates. Multi-turn baselines conduct active interviews via an oracle-simulated user under a 15-turn harness cap, prompted to conclude within five to eight questions: `gemma3:4b` [Google DeepMind, 2025b], `llama3.1:8b` [Meta AI, 2024] and Gemini 2.5 Flash [Google DeepMind, 2025a] on $n = 50$ prompts, and Claude Haiku 4.5, Sonnet 5 and Opus 4.8 [Anthropic, 2026], two of them on the full 150. Three asymmetries of this protocol favour ALI, and we state them before any figure: the baselines were instructed to stop within eight questions while ALI receives no instruction about length; the baselines' simulated user may answer "I'm not sure" where ALI's must invent a plausible concrete value; and ALI reads the parameter list the scorer marks against, which no baseline sees. We measure the first (Section 4.4) and declare the other two (Section 5); Supplement § A gives all four prompts verbatim. *ALI implementation*: the same 150 prompts run through the full pipeline (C0–C4) as **ALI_v1** (C1 fine-tuned, C4 Gemini), **ALI_v2** (full Gemini) and **ALI_v3** (C1 and C4 fine-tuned). This harness gives C3 a single candidate per turn from the highest-weight cluster; the six-candidate reward ranking belongs to the deployed systems, which the ablation below suggests matters little.

Reproducibility. Every figure comes from one scored artefact per configuration: we store the prompts, the weighted parameter spaces and the resolved frame F of each run together, so any coverage value recomputes as $\sum_{p \in F} w_d(p) / \sum_{p \in \mathcal{P}_d} w_d(p)$ without a model call, and all 150 ALI_v1 runs return the stored value exactly. Bootstrap intervals use $B = 10{,}000$ resamples at seed 42; re-executing an unchanged configuration two weeks apart moves a domain mean by 0.9 pp (37 of 50 runs identical), the resolution at which to read differences here. Code, prompts, taxonomies, stored runs, the annotation sheet and the supplement (`supplement.pdf`) are released at https://github.com/aonymous412/interview-pipeline-release (Supplement § A).

4.2 An Interview, Turn by Turn

Table 1 replays one Consulting episode of ALI_v1 from the stored run. C0 opens the interview at 34.6% from the initial message alone; the first question resolves two parameters at once, each later one resolves a single parameter in strictly descending weight, and the loop halts at turn 8 the moment $\mathrm{cov}(F) \geq 0.90$ holds, leaving four low-weight parameters (90 of 978) unasked. On the same prompt, Claude Opus 4.8 asks reasonable questions the parameter list does not weight (custom-cake lead times, delivery radius, loyalty mechanics, time slots, staff dashboard) and declares itself done after six at 51.4% coverage, budget, timeline, design and branding never raised. Neither interviewer is wrong about what a bakery needs; only one is measuring what remains.

4.3 Coverage Under an Explicit Criterion

Table 2 collects every system we evaluated on one task, under one scorer; its three groups differ on a single axis, how the system decides it has heard enough. The first two groups stop when the model declares itself done, under the five-to-eight-question instruction of Section 4.1; the third stops when $\mathrm{cov}(F) \geq \theta_d$ holds, with no instruction about length. Every comparison is paired: the $n = 50$ rows use a strict subset of the 150 prompts, so no row draws an easier sample.

Single-pass extraction establishes the floor at 40.7% [37.5–43.9%], with a consistent domain gradient (Consulting 48.9% > Medical 41.7% > Payments 31.5%). Down the middle group, coverage climbs from the 4B local model to Gemini 2.5 Flash and then flattens: the three frontier

Table 1: ALI_v1 on prompt `P12`: "I want an e-commerce site for a French bakery. It should support online ordering with delivery and pickup options, a loyalty points system, and Stripe for payments." Consulting, $\sum w = 978$, $\theta = 0.90$; answers in Supplement § E.

t	Question (C3)	Resolved by C4 (w)	cov
1	What are the main pages or sections you envision for the site?	pages_structure (90) + user_accounts (40), volunteered	47.9
2	What visual style, aesthetic direction, and overall branding are you envisioning?	design_style (75)	55.5
3	Do you have all the content and copy ready, or will we need to create it?	content_ready (70)	62.7
4	Do you have an existing logo, specific brand colors, or any brand guidelines?	existing_branding (65)	69.3
5	What are your preferences for the e-commerce platform or technology stack?	technical_platform (60)	75.5
6	What is your estimated budget range for this project?	budget_range (55)	81.1
7	What is your desired launch date or timeline?	timeline (50)	86.2
8	Do you have any specific SEO goals, domain preferences, or search visibility strategies?	seo_requirements (45)	**90.8**

Turn 0: C0 pre-extracts purpose (100), audience (95), features (85), integrations (58): cov = 34.6. *Stop at turn 8:* $\mathrm{cov}(F) = 0.908 \geq \theta$; unresolved content_management (35), analytics_needs (30), multilingual_support (15), accessibility (10).

Table 2: All evaluated systems, one task, one scorer, grouped by how termination is decided. Calls = model invocations per interview; tokens = input plus output, simulated user excluded (provider counters for ALI and single-pass; baseline rows reconstructed with one shared tokenizer, ≈12–20% below the provider's count). 95% bootstrap CI (B = 10,000, seed 42).

	System	n	Mean cov.	95% CI	Turns	Calls	Tok. (k)
Single-pass	Gemini 2.5 Flash (frontier)	150	40.7%	[37.5–43.9%]	1	1.00	3.9
Multi-turn LLM, no formal stopping criterion (instructed to stop within 5–8 questions)							
	`gemma3:4b` (4B, open)	50	42.2%	[37.2–47.2%]	4.2	4.18	2.0
	`llama3.1:8b` (8B, open)	50	47.5%	[42.7–52.3%]	5.9	5.90	3.0
	Gemini 2.5 Flash (frontier)	50	52.8%	[48.5–57.2%]	7.0	7.02	3.6
	Claude Haiku 4.5 (frontier)	150	57.1%	[54.4–59.9%]	6.43	6.43	4.9
	Claude Sonnet 5 (frontier)	50	59.9%	[55.1–64.6%]	6.86	6.86	4.7
	Claude Opus 4.8 (frontier)	150	58.4%	[55.5–61.2%]	6.75	6.75	5.2
ALI, termination governed by $\mathrm{cov}(F) \geq \theta_d$*, no instruction about length*							
	ALI_v3 (C1-FT, C4-FT)	150	51.6%	[46.4–57.0%]	12.86	36.80	15.6
	ALI_v1 (C1-FT, C4-Gemini)	150	**89.0%**	**[88.2–89.9%]**	**6.93**	**19.90**	**7.8**
	ALI_v2 (full Gemini)	150	89.4%	[88.5–90.3%]	6.99	20.60	9.4
	ALI_v1, uniform $\theta = 0.90$	150	92.1%	[90.9–92.8%]	7.36	21.10	8.2
	ALI_v1, 8-turn cap (re-analysis)	150	84.4%	[82.8–85.9%]	5.95	—	—

models land within 3 pp of each other despite large capability and price differences. Under the formal stopping rule, ALI_v1 reaches 89.0% [88.2–89.9%], +30.6 pp over the strongest baseline (Claude Opus 4.8, 58.4% [55.5–61.2%], same prompts, same scorer, disjoint CIs) and +26.0 pp at that baseline's question budget. The stopping rule does the work it is meant to do: 146 of ALI_v1's 150 runs halt on $\mathrm{cov}(F) \geq \theta_d$ (97.3%; 96.7% for ALI_v2; 30.0% for ALI_v3). Per domain (Supplement § C), the margin over the strongest baseline is +35.4 pp in Consulting, +27.0 pp in Medical and +20.4 pp in Payments, with disjoint intervals in all three; at the baselines' eight-turn budget, +30.1, +18.4 and +20.3 pp, still disjoint; the strongest generalist changes with the domain. Payments halts after 3.98 turns because $\theta_d = 0.75$; it is also the domain where the scorer agrees least with human annotators ($\kappa = 0.539$).

The last two columns price that gain. ALI_v1 spends 7.8k tokens per interview against 5.2k for Claude Opus 4.8 (≈6.0k after the reconstruction factor), a ratio of 1.3 to 1.5, over 19.9 model calls instead of 6.75; per point of weighted coverage the ordering reverses (88 tokens against 89 to 102), and the tokens are cheaper ones. A baseline resends the entire conversation at every turn, so its input grows quadratically at frontier prices; ALI issues short, single-parameter requests on Gemini 2.5 Flash and on 124 M–1.5 B open models. It buys coverage in calls rather than in context, which is also why a frontier provider has little incentive to adopt it (Supplement § D).

Capacity does not explain the plateau. Claude Opus 4.8 terminates on all 150 prompts at 58.4%, 6.75 turns on average; it was instructed to conclude within five to eight questions and no run exceeded eight, so this plateau sits under a bounded turn budget rather than at a freely chosen stopping point. Two controls separate the budget from the stopping rule: ALI_v1 spends 6.93 turns on average, within a quarter-turn of the baseline, and truncating ALI_v1 to the same eight-turn budget still leaves +26.0 pp (last row of table 2), so the budget accounts for 4.6 pp of the gap. The two other asymmetries of Section 4.1 are not measured here (Section 5), so the 26.0 pp that survive the budget control are an upper bound on the criterion's own contribution. Baseline coverage rises with prompt specificity (52.9% ultra-vague, 57.4% medium, 65.0% detailed), as expected if the input rather than the model binds. Model capacity shifts this plateau by single digits at a fixed turn budget; imposing $\mathrm{cov}(F) \geq \theta_d$ shifts it by 26 pp at the same budget and by 30.6 pp unconstrained.

ALI_v2 ≈ ALI_v1. The full-Gemini configuration achieves 89.4%, not significantly different from ALI_v1 (overlapping CI): the formal stopping rule and taxonomy-driven loop account for the gain and the PEFT adapters contribute nothing measurable; the present runs do not separate the rule from the taxonomy it reads.

Resolved versus mentioned. Coverage scores whether a parameter is *resolved to a deployable value* ("volume per trade: $200"), not merely *addressed* ("you should define your per-trade volume limit"). Across the five highest-weight Payments parameters, single-pass Gemini extracts a concrete value in 33–60% of responses and addresses the rest without resolving them (Figure C.1, Supplement § C); for safety-critical enforcement the distinction decides outcomes, which is why FinAgent's deterministic engine accepts no "mentioned" parameter in place of a "resolved" one.

4.4 Ablation

Architectural commitments that matter. Three ablations move the result, all three in table 2. *Removing the stopping rule while keeping the model.* Gemini 2.5 Flash interviewing on its own judgement reaches 52.8%; ALI_v1, which runs that same Gemini behind C4, reaches 89.0% on the same prompts under the same scorer. We hold the model fixed and vary the architecture, which isolates the architecture (criterion, taxonomy-driven loop and their simulated user) from the capability of the model underneath it; it does not separate the criterion from the taxonomy the loop reads, a condition this paper does not contain. *Degrading C4.* A fine-tuned sub-1B extractor (ALI_v3) collapses coverage to 51.6%, inflates the interview to 12.86 turns and drops the share of runs halted by the criterion from 97.3% to 30.0%: the frame stops growing and the loop runs to the cap. A formal stopping rule is only as good as the component that advances the frame. *Constraining the turn budget.* Truncating every ALI_v1 run at the baselines' eight questions and rescoring the partial frames gives 84.4% [82.8–85.9%] at 5.95 turns, still +26.0 pp with disjoint intervals and fewer turns than the strongest baseline spent.

Components the result does not need: a negative result. Five variants (full pipeline; no clustering; no priority ordering; C1 random ordering; C3 template tier only) run on a paired subset of $n = 30$ prompts, 10 per domain (Supplement § F, Table F.1), and land between 89.9% and 90.4% coverage and between 6.1 and 6.6 turns. No variant differs significantly from the full pipeline on either metric (Wilcoxon signed-rank on paired differences, Holm–Bonferroni; all adjusted $p = 1.000$), and two one-sided tests with margins fixed before the run (2 pp of coverage, one turn) establish equivalence for all eight comparisons ($p < 0.004$). The design is powered to detect 0.8–1.5 pp and 0.7–1.0 turns, well inside those margins, so this is a demonstrated absence of effect rather than an inconclusive test, and it contradicts simulation-derived predictions of a near-doubling of turns without clustering. These variants degrade question selection and leave coverage tracking in place; Supplement § I shows that coverage-driven selection carries no entropy-optimality property for them to degrade. The result therefore rests on the pair no variant removes: an explicit coverage criterion, and a C4 able to turn an answer into a resolved parameter. Clustering, weighted ordering, tiered generation and the PEFT adapters are design choices, and we report them as such.

5 Discussion

The reported gap should be read as an upper bound on what an explicit stopping criterion contributes. The multi-turn baselines were instructed to conclude within five to eight questions, their simulated user was allowed to answer "I'm not sure" where ALI's was instructed to invent a plausible concrete value, and ALI alone reads the parameter list the scorer marks against. We measure the first asymmetry: holding ALI to eight turns keeps +26.0 pp of the +30.6 pp (row *8-turn cap* of table 2). The other two are untested, and settling them requires a baseline re-run under ALI's oracle and a condition that places the weighted parameter list in the baseline's own prompt, neither of which this paper contains.

Every system in table 2, ALI included, is evaluated against a simulated user [Schatzmann et al., 2007]. Simulated users are more cooperative than real ones, which inflates coverage for all systems and need not do so equally; the real-user protocol is designed (Supplement § H) and has not run.

The three domains, their parameter lists and weights, the prompt set and the scorer are all ours. The scorer is validated against 516 human annotations and we control for the known biases of an LLM judge [Zheng et al., 2023], but no external benchmark anchors the task; evaluating the criterion on a public intent-elicitation benchmark such as IN3 [Qian et al., 2024] is the step we would take before generalising.

On Payments the scorer misses its pre-registered agreement threshold ($\kappa = 0.539$), and a second judge from another model family misses it too. We read this as ambiguity in the taxonomy rather than as a scorer defect, and note that dropping Payments widens the gap to +35.8 pp rather than narrowing it.

Little of the result is learned. The all-API configuration matches the one with fine-tuned adapters, the paired ablation (Section 4.4) finds clustering, weighted ordering and tiered generation equivalent to simpler choices, and the fully fine-tuned configuration collapses to 51.6% because a sub-1B extractor cannot advance the frame. What carries the result is an explicit criterion and an extractor good enough to serve it.

The approach applies wherever a downstream pipeline consumes a structured frame and an expert can write a weighted parameter list in one to two days; it does not apply where the parameter space cannot be enumerated in advance, and errors in extraction propagate directly into the frame the criterion trusts. The full list of limitations is in Supplement § K.

6 Conclusion

We formulated user intent understanding as weighted coverage optimisation, instantiated it in a five-component pipeline and validated it across three applied systems; the coverage metric (Eq. (1)) and objective (Eq. (2)) give single-pass extraction the explicit completeness criterion it lacks. On $n = 150$ prompts under a validated scorer ($\kappa = 0.749$), single-pass extraction reaches 40.7% and the strongest multi-turn baseline (Claude Opus 4.8) 58.4%, three models of one family within 3 pp of one another; ALI_v1 reaches **89.0%** [88.2–89.9%], +30.6 pp (+26.0 pp at the baselines' question budget), with 146/150 runs halted by $\mathrm{cov}(F) \geq \theta_d$, and ALI_v2 (full Gemini, 89.4%) performs on par, locating the gain in the stopping rule and taxonomy-driven loop rather than in the adapters. A broader generality claim needs independent replication.

Perspectives. Four steps follow from Section 5, in order: (1) the baseline re-run it describes, which settles how much of the gap the criterion alone explains; (2) real-user validation across the three deployments; (3) an external anchor on a public intent-elicitation benchmark; and (4) testing on real taxonomies the comonotonicity condition of Supplement § I, under which coverage-driven and entropy-based selection agree.

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